

"Physics as Information Processing" ~ Chris Fields ~

Lecture 6

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2023 **Duration:** 1:06:48 | **Uploaded:** Unknown **Transcript method:** auto_caption

hello and welcome everyone it is October 12th 2023 and we are in the final lecture session six in physics' information processing with Chris Fields so Chris thank you for this journey exciting to get to this final lecture which will be followed only by the the last discussion for this go around so it's really awesome and to you for this lecture uh thank you Daniel and uh welcome everyone to this uh last session that I'll be doing uh with a focus on biology and just to give A Brief Review as as usual uh We've focused in this course on generic Quantum systems uh by which I mean just any Quantum system uh that has some number of degrees of freedom and we really haven't made any assumptions about these systems other than the assumption that both interacting Partners have sufficient degrees of freedom that um they can be considered separable uh so their boundary is small with respect to the total size of each of the inter interacting systems and I want to emphasize again that that the approach we've taken is completely topological so we haven't assumed anything about embedding in space um and as we've characterized these systems we focused really on two things uh one the N the amount of information that the two systems can exchange and that amount is determined by the size of their boundary it's determined by this number in which is the Hilbert space dimension of the boundary and as we've discussed now in several of the sessions we can think of this boundary as a collection of in cubits via which the two systems A and B communicate the other uh feature of the interaction on which we focused um is the reference frames that are used by the two systems to interact with this set of cubits and it's these reference frames that assign the information that flows between A and B uh meaning or functional significance or actionability or however you want to uh refer to the idea that uh when a system gets some information it does something useful with it so those are very generic kinds of ways of characterizing an interaction uh you can think of it as a qualitative characterization the the semantics of the information and a quantitative characterization the amount of information but other than that we've left the the nature of these systems open and this characterization allowed us to

reformulate the free energy principle in uh Quantum information theoretic terms as the principle that interacting systems behave in a way that will asymptotically align their reference frames and we saw that full alignment of reference frames uh is equivalent to entanglement so the interact the the FP drives systems toward entanglement which is not surprising because that's uh what happens to generic Quantum systems they may start out separable but if they interact long enough they'll become entangled so we've had a discussion that is really very uh generic now we showed a couple of sessions ago that under these very generic assumptions we got some interesting outcomes and one is that if systems have sufficient degrees of freedom uh under the action of the Fe they will be driven toward making um a diverse set of measurements of their environment because they're trying to get information about it so that they can predict what it's going to do and that as they add ways of making measurements they'll come up against our non-commutativity of their measurements and this may be due to Intrinsic mathematical facts about their measurements and it may just be due to their limited supply of free energy so I'll either way what this induces is compartmentalization and uh the requirement for internal classical communication on internal boundaries so I just wanted to to reemphasize that this is a result of basic physics and we see compartmentalization of course ubiquitously in biology um but this suggests to us that uh features of biology that are very familiar actually follow from from Fairly deep physics and compartmentalization is one of them we then talked last time about space and time and the computational requirements of on any system that is able to perceive an external clock or perceive an external spatial layout and we discussed this these uh from a from a very kind of Developmental Psychology point of view uh we discussed it using essentially the kinds of terms on the right here as opposed to uh the kind of language that's much more Technical and mathematical that's used in physics to uh derive SpaceTime as emergent from information transactions but we at least hypothesize that both of these are ways of getting to the same sense of emergent SpaceTime and one important outcome of this discussion was that systems can perceive external clocks so external periodic Behavior with fewer computational resources so before they're able to perceive spatial layouts and we'll get back to this later so organisms and maybe other complex systems seem to manage um a collection of uh capabilities that are all mutually dependent so um memory depends on having layouts of persistent things that can be distinguished uh so it depends on space and it certainly depends on time um space gives structure to persistent objects memory allows you to identify persistent objects and objects in turn give structure to space by being located in different places and objects enable the fundamental uh requirement

for memory which is some form error correction so that one has enough redundancy in the memory to check to see whether a memory has been degraded so organisms um seem to manage uh a physically very important process which is the use of SpaceTime and memory as a basis for cognition uh you'll recognize these four functions that are that are mutually interdependent as underlying complex cognition across the board and you can't really do complex cognition unless you have these capabilities so today we're going to put some of these things together in a discussion of biology and and in by looking forward to Future directions not just in biology but in all of quantum information Theory so let's talk about biological systems living systems uh which of course includes us uh but also includes all other organisms and communities of organisms and so forth at multiple scales and I want to put forward a fairly um radical hypothesis that living systems are just just generic Quantum systems that the Fe has driven to a high level of computational complexity so what does this belief say it suggests that living systems are just Quantum systems that have enough degrees of freedom and a sufficiently complicated internal dynamics that the Fe by requiring them to predict what their environment is going to do next has driven them to make complicated measurements and complicated inferences on the basis of those measurements that enable fairly complicated actions on the environment so these are systems that you would characterize as computers essentially information processing systems with a high degree of capability now this is is clearly uh as much a philosophical hypothesis as an empirical hypotheses and uh that's due in part to the fact that we don't really have a good definition of what a living system is we can just point and say you know that's a living system and that thing over there isn't uh but beyond that we don't have much of a theory and of course people have worked on a theory of what is life for decades um but there's not a lot of agreement and so this is uh a in a sense a definitional kind of hypothesis and we test such kinds of hypotheses by seeing if they're useful conceptually to help us understand what living systems are doing now soon as we as we put forward something like this inevitably the question comes up are living systems really Quantum systems or are they classical in some intrinsic sense and this question is really an artifact of of two things one the continuing association between Quantum systems and microscopic systems uh which is of course traditional quantum theory was developed to deal with microscopic phenomena it was first tested uh as a theory of microscopic phenomena and it's only in the past couple of decades that quantum theory is being applied to big things uh to cosmology for example so uh there's there's a long tradition of association between Quantum and microscopic and living systems are mesoscopic um much

bigger than atoms bigger than molecules so and they're not nearly as big as cosmological systems so they're somewhere in between and those systems have traditionally been regarded as classical and even chemistry even biochemistry is still taught in a very classical way so we're not used to thinking of these things as Quantum systems but at least in the approach we've taken in this course um all physical systems are quantum systems and classical information exists only on boundaries only on boundaries between systems and it only describes their interaction and I want to emphasize I mean know I've said it before but since this is the last class I want to say it again that this is not a universally held view um it's it's a view that's becoming more common um and I think a generation from now it will probably be the dominant view of things uh but it's it's not fair to say that it's the dominant view now it's it's I think quite fair to say that it's a view on the rise especially in the quantum information and quantum gravity communities but from a strict empirical perspective uh we know a number of things uh we know that accurate molecular Dynamics calculations require quantum theory uh we know that a lot of uh biologically important molecular processes uh use quantum coherence we know that things like hydrogen bonding can only be understood uh using quantum theory uh we know that cellular free energy budgets are way too small to support fully classical information processing if we think of information processing in terms of what proteins are doing or what other molecules are doing in terms of changing shape know we know that shape changes in in proteins and other molecules underly all cellular information processing so if we think of them as actually implementing cellular information processing we have way more information processing than could be funded classically and finally we know that human cognition exhibits contextuality so human cognition uses Quantum coherence it's not a classically a classical causal process in all cases but this is still a very new field and it's a field with a lot of potential and I think again a generation from now uh Quantum biology is going to be regarded as a very mainstream Endeavor uh even though it's something of a small emerging field now and I'll just show you one picture uh in support of this third Point um these are this is a plot of uh measured uh free energy usage converted to numbers of bits processed at one kilohertz for a number of of procaryotic and eukaryotic cells uh all that I could find a few years ago that had good uh calorimetric measurements versus the log of their area their uh membrane surface area and the thing to note here is that these cells are processing classical information at pretty low rates kilohertz to gigahertz rates for a thousand bits at a time and we know that a lot more is happening uh in cells than that now cells have many more than a thousand proteins and those proteins are all moving all the time and if that

motion is processing information then there's not nearly enough uh classical not nearly enough thermodynamic power to power all of that information processing so let's now talk a little bit about life and living systems and uh what they look like now here's a view of living systems from the point of view of genome diversity this is a few years old so it's out of date by now but it tells us a lot qualitatively uh it tells us that almost all genome diversity is microbial that the sorts of organisms that we're familiar with medium sized organisms like us animals and plants have only a tiny amount of genome diversity so we think of ourselves as pretty diverse we think of ourselves as quite different from other animals from flies or worms or sponges or something like that but in fact we're not all that different from them we share many many proteins with all of those other organisms we share the same cellular organization and so forth we're not in fact all that different from plants and uh we're we're much more similar to the fungi than to the plants so we get a very biased view of genome diversity when we just look at familiar large types of living systems uh most genome diversity is buried down there in the microbial world this isn't the only way we can look at life uh as a diverse set of organisms we can also look at it in terms of a cell lineage uh the fact that we're all related at the DNA level tells us that we're all descendants of a common ancestor which is usually called Luca the last Universal common ancestor and uh Luca is generally modeled as being a cell uh more or less like a bacterial cell so fairly simple uh with not very many pathways uh with maybe a DNA genome and if Luca didn't have one then they arose fairly shortly thereafter because all organisms that we know of have DNA genomes but the point of looking at life as a lineage is that we get a diagram that looks very much like the cell lineage in from an embryological situation much like the cell lineage of our own bodies growing from a fertilized egg and in particular looking at life as a lineage shows us that not only do we share DNA with all other organisms we actually share a membrane cytoplasm that's continuous in time all the way back to Luca and so continuous in time between us and bacteria so if you look at life this way you realize that that life is in a sense all one organism all one connected structure that's developed over time through division cell division and so we are very closely related uh to all other parts of life in the same way that different cells and our bodies are closely related to each other the time span just longer I mean here the time spans three and a half or four billion years uh between us and Luca these lineages have been diverging for that long but they're still continuous and in particular uh they form a continuous cell membrane so when you think of life in this way we realize that we are reasoning from just one example when we think about terrestrial life and this strongly limits what we can say about

living systems what we can say about biology that we really only have one example the the lineage that descends from Luca here on Earth and at some point we may have more examples and there's certainly groups working on artificial organisms that aren't part of this lineage that are constructed from Parts derived from this lineage but aren't members of this lineage themselves and that may tell us something new about life but if we if we think of life this way um we see something very important which is that as life grows its boundary gets bigger and since the boundary of a system constitutes its informational environment uh its Dimension getting bigger means that more information flows into it from its environment and more information flows out of it into its environment so as as life develops from Luca it gets a larger and larger informational environment and this Convention of its boundary grows and grows now what that means if you go back to our original picture of two interacting systems is that the number of the dimensionality of the physical environment is decreasing uh and we think of the physical environment as increasing in size we have the expanding universe and all that but what's increasing in size is space and if space is emergent then that increase is not a real increase in degrees of freedom what we're really interested in is the Hilbert space of the physical environment as distinct from the Hilbert space of life and as life gets bigger it it literally quite literally eats the physical environment and turns it into more life so there's less physical environment around so the trajectory of life is toward a bigger information AAL environment and a smaller physical environment and as the physical environment gets smaller you might imagine it gets easier to predict as the informational environment gets bigger you have more information with which to predict the behavior of the physical environment so this all hangs together very nicely with the free energy principle here's another picture of this same thing um if we start with some boundary here on the left lab B around some system labeled s and it's embedded in an environment e then as s makes copies of itself um through some process like cell division then it starts to fill up more of e it's it's actually converting degrees of freedom of e into degrees of freedom of its copies that it's made of itself and if we think of some larger boundary drawn somewhere in E here on the right side I've avbl that b Prime what happens is that b Prime gets slowly filled up with copies of s and it starts to look like a large organism and that's what life is doing in the physical environment it's making a bigger boundary for itself at the expansive degrees of freedom of its physical environment so we can ask what drives this process uh why is life doing this why is it making copies of itself and the answer to that question at least one answer to that question is the free energy principle drives it to do this and uh I'll just show you some simulations uh from several years ago of a cell

population uh in this upper left panel the the green cells are stem cells that divide and they've been eded in an environment that's safe on the left side and 100% lethal on the right side so this they're embedded in it in an environment that's that's conducive to life on one side and very dangerous as you get toward the middle and the purpose of these um simulations was to study what happens when the stem cells are able to divide and create not just other stem cells but create specialized progeny that don't divide so sematic progyny U that don't divide and hence have free energy resources thermodynamic resources that they can devote to doing something other than dividing cell division is incredibly expensive when it comes to energy so if you take that energy and apply it not to cell division but to protecting the stem cells um you can allow the stem cell population to expand and that's what these other simulations show the blue cells are sematic cells that have very varying ability to protect the stem cells and if they're fairly good at protection and they're fairly hard to kill uh down in the the right bottom quadrant um you can actually get a population expanding well into the highly lethal part of the so this shows you the Fe in action uh what the stem cells are doing is constructing a very predictable neighborhood around themselves in the environment by dividing to make progyny that don't themselves divide that just um sit there and and help Pro protect the stem cells so the stem cells are now living in a safer more predictable environment and that allows them to invade more and more of the of their physical environment by having this fairly benign informational environment so this is the FP in action driving the expansion of Life at the expense of the physical environment so that lets us um go back to this question that was raised last time about the mechanism of these scale transitions that we see ubiquitously in life uh life gets bigger and bigger um as time goes on F genetically and as time goes on developmentally uh we start with some molecular Pathways that somehow get assembled into a cell that's the origin of Life question how does that happen and about two billion years later cells start to group together to make multicellular organisms now bacteria group together to make facultative multicellular systems like microbial Maps three billion years ago very very early on uh but we don't see the emergence of obligate multis cellulars until much much later uh before the Cambrian explosion but still less than a billion years ago and from there things get more complicated very quickly so uh one has large multi cellular organisms uh they develop complicated social relations that probably first emerged uh with social insects uh but are certainly welldeveloped in in vertebrates including in mammals um organisms uh group together uh in very complex ecosystems and um recall here that it's now known that all multicellular organisms have obligate microbial symbiance so all of our bodies are in fact very

complex ecosystems populated by many many different species of bacteria and that's true not just for us but for trees and insects and worms and sponges and everything else so um organism from a from a body point of view are themselves ecosystems and organisms organize themselves into ecosystems and these ecosystems come into contact uh across the entire biosphere so um process is as simple as the wind uh disperse organisms as small as bacteria all over the planet so this scaling of Life toward larger and larger organizational structures is in a sense the major thing for biology to explain and the hypothesis that we floated last time was that this um transition from very small scales to very large scales uses the same mechanism over and over again or a a mechanism that has the same structure over and over again so the the renormalization group is is Trivial basically it's this it's just the identity operating at different scales or mapping from one scale to the next so the question is can we can we approach this in any way uh does this make sense and to to try to make sense of it um we can go to the the one real developed theory that we have in biology which uh has been formulated by Dawkins and others as universal Darwinism uh this is just kind of an abstraction of the the modern synthesis uh between darwinian evolution and genetics but Universal Darwinism is incredibly simple it it's and very obvious it says if you have systems and they can copy each other they can copy themselves and the copies can diversify then if you want to know what's going to happen at the population level uh what you will find is the systems that copy themselves and diversify most efficiently will win because there'll be more of them so they end up dominating the population and uh winning gets called natural selection but natural selection is not an extra part of the darwinian process it's just the outcome of efficiency of the systems that copy themselves most efficiently coming to dominate uh any population so there's no sort of um kind of hand of God or whatever reaching down and selecting some organisms and not others and of course when Darwin put the idea of natural selection together he was he was thinking very explicitly of the analogy of of a gardener or a farmer uh selecting the the crops or the animals or whatever that that the gardener liked best and keeping those and that's where this selection metaphor comes from but there's no real process selection in uh evolution by natural selection it's just copying and diversifying but this clearly is just the outline of what a real mechanistic Theory would be because we need to to have a story about what copying is and what diversifying is and what efficiently means um and in particular to address the question that we raised a couple of slides ago we need to know whether these mechanisms are the same or at least isomorph morphic of the same form at different scales so um that's what I want to talk about for the next little bit let's start with copying um up until the

70s copying in biology really meant DNA replication and to some extent cell division but the focus in evolutionary theory was on DNA replication and in the 70s uh it was realized that whole genes could be duplicated so whole genes could be copied and then shortly uh after that it was realized that particularly in procaryotes where genes functionally related genes are next to each other in chromosomes often in what are called operons that operons could be duplicated and so entire Pathways could be duplicated and as genomes began to be sequenced and structures of genomes were understood it became clear that genomes could be copied and uh of course we have two copies in each of our cells with the exception of red blood cells of our complete genome but some plants for example have six or eight or even more copies of their genome in each of their cells so there's large scale genome duplication uh cell division is clonal duplication of cells organismal reproduction works by cell division in all known systems um colonies or societies of organisms reproduce themselves in part through cultural transmission and entire niches reproduce themselves uh for example in processes like um Forest Evolution or desertification or even urbanization uh we see humans reproducing the same sorts of niches in city after City for example so these forms of copying um have the kind of hierarchical dependencies that one might expect so to duplicate Pathways you have to duplicate genes uh to duplicate cells you have to duplicate The genome to duplicate organisms you have to duplicate cells and this suggests that this copying operation is indeed scale free that that each each level the copying operation at each level enables the copying operation at the next level now with diversification we see much the same thing um in the early days of the modern synthesis and and still in the work of people like like Richard Dawkins diversification is is generally thought of as random so mutation for example is thought of as random and um here I have Barbara Mento who is kind of the hero of non-randomness in um doggedly over decades pointing out that much of genome diversification is not random at all and [Music] um that genomes actively diversify by moving whole clusters of genes around and uh duplicating them or just changing their positions so that they're under the influence of different local uh control so at the Gene and genome level uh you have active diversification by various processes uh entic processes or as as Mento showed us transpose on mediated mechanisms which are enzymatic at the at the bottom um cells actively diversify by lots of different mechanisms and uh in the 70s uh Carl W was in part large personally responsible for pointing out that in the microbial world there's a lot of Gene transfer and uh again no one believed him for quite a oil but now it's it's broadly accepted that microbes exchange DNA uh very frequently and across large distances phenetically and this is how things like microbial

resistance to antibiotics spread so quickly or one way that it spreads so quickly and it's a tremendous agent of essentially speciation to the extent that that even makes any sense in the microbial world of course in our cells we have functional differentiation uh as mentioned earlier there are lots of symbiotic interactions that lead to diversification in cells by radically altering their local environments and of course sex is a mechanism for diversification uh organisms actively diversify in many different ways the most obvious one is just by moving around by moving to a different environment uh the organism is exposed to different inputs and is allowed outputs and so it can move into a different direction U morphologically genetically Etc uh organisms can regulate their own evolvability by altering things like DNA repair systems or activating transposons again these are sorts of things that uh went very much Against the Grain when they were first discovered but are now pretty well accepted and of course organisms can learn and so can diversify their behavior by diversifying their capabilities uh and social groups of course actively diversify uh in many ways and uh organisms are increasingly non-human organisms are increasingly regarded as having cultures that they pass on intergenerationally by intergenerational learning and can exchange with with other groups so diversification uh is active and occurs at every scale and again diversity Al although this is a bit less clear than in the case of coping uh diversification mechanisms at one scale enable diversification mechanisms at other scales so for example uh transposon mediated mechanisms uh enable horizontal Gene transfer in in microorganisms and e f tells us that efficiency is actually about intelligence because the FP drives systems to build good models of their environments or the behavior of their environments and what a good model lets a system do is reach a single goal by many different means because it allows in a sense planning um and it allows flexibility and adaptability and I put William James here on this slide because he was the first as far as I know to Define intelligence as this ability to use different means to reach the same goal now in evolution that's that's long been known and it's called convergent evolution when you see two lineages adapting to a single environment with very different genetics and very different morphologies and uh often the organisms will end up with similar morphologies that work well in that environment but very different genetics and very different biochemistry so throughout Evolution we see this phenomenon of functional convergence balancing structural Divergence at many many different levels so we have multiple genes for very similar proteins we have multiple pathway ways that do more or less the same thing uh we have different cell types that support the same basic functions uh we have tissues that have that are are fairly different uh but that have the same functions for

example eyes and different kinds of organisms that allow Vision but have very different structure and of course we have functional convergence and social relations and if we look at how cells interact um traditional evolutionary theory really focuses on competition but we now know that cells cooperate that they exchange resources in various ways uh even across vast differences in in phylogeny for example The Interchange between plants and fungi or between bacteria and us um we know that cells in a sense persuade each other by uh secreting things like hormones that cause other cells to do different things um cells deceive each other uh for example microbes that are trying to escape the immune system uh and cells coers each other cells even kill each other when they uh detect for example genetic defects so these sorts of of Social and economic and even political uh ideas are useful for describing living systems from the level of cells on up I mean they were developed in Social settings uh to describe what goes on in human societies and then extended to animals and now it's very clear that they extend to individual cells so when we think about social intelligence we can think about social intelligence even at the level of individual cells and we see these sorts of of competition cooperation resource exchange Etc even at the level of microbial Mass so biological structures that are three billion years old now of course intelligence is necessary but it's not sufficient for being selected it won't protect you from impacts of of asteroids that happen to land in your planet uh which has happened often in the evolution of the earth and we're now trying to figure out how to predict and and prevent such things uh but it's you know who knows whether we'll be successful with that uh but and this intelligence is necessary when it's viewed in this Fe sort of way intelligence is what lets you build models that um enable you to predict the environment so where does this leave us um by thinking of evolutionary theory in terms of scale-free mechanisms we seem to have a picture in which we start with Quantum systems with a certain level of complexity and the FP actually gives us scale-free biology or something very close to scale-free biology and that's very interesting because the FP of course is a generic principle it just says that systems do what they need to do to keep existing and this way of thinking gives us some predictions uh one is that we'll find compartmentalized problem solving everywhere uh because the FP drives compartmentalization and uh the reason is just non-commutativity of of measurement processes it predicts that we'll see hierarchical organization everywhere uh in the same way we did on that first compartmentalization slide where once you have a number of compartments you need a meta compartment that if nothing else distributes free energy so some sort of very basic attention system and we see attention systems of course throughout biology um it predicts that living systems will have very limited

classical data structures so since they have to write classical data on their boundaries uh and their boundaries are much smaller than they are then they'll and because writing classical data is expensive that living systems will tend to use as little classical data as they can get away with since classical data is needed for thermodynamic irreversibility where we see thermodynamic irreversibility uh that's where we look for classical data uh we saw that organisms will be able to detect time first uh before they can detect space so we can expect organisms to uh have clocks before even before they can uh represent space and of course we see things like dial variation throughout biology and finally the the FP predicts this asymptotic collapse of separability back to entanglement so it predicts that our living systems will eventually degrade and become indistinguishable from their environments and we know that happens at the cell level and the individual organism level uh you know we call it death so the Fe seems to give us a lot of the features of biology and it gives us these features in a scale-free way and it derives them from this very basic physics so where does that leave us I think it leaves us here with a challenge not just to thinking about the Fe which is what this course has been about but to Quantum information theory in general and that challenge is to evolve from the kind of conceptual and theoretical utility that we've been discussing in this course and which is very well developed in the physics Community to of some sorts of empirical and technological utility and I've listed three areas here where qit is very active uh one in Quantum gravity and cosmology and you're probably familiar with things like string theory that sort of fairly famously seem not to be empirically testable and the quantum gravity Community is enormous uh many thousands of people working in this area but coming up with theories that can be tested empirically uh at the scales that we can actually do experiments so at accessible energy scales for example has proved to be very difficult um so this is a real challenge to the field is somehow uh getting not just to empirical testability but to some sort of technological utility and you can imagine that if the the main prediction of quantum gravity is that SpaceTime is emergent that um there could be enormous utility in that uh some ability to manipulate the way SpaceTime emerges the second area of course is quantum information and Computing and here massive resources are being put into uh technological utility and I think it's reasonable to expect that we'll have useful quantum computers within a generation at any rate so there I think the the field is going quite well and the last area what we've talked about here is quantum biochemistry and biology and this I think is an area with enormous practice promise that um maybe some of you will actually contribute to uh making this area more useful technologically and more accessible empirically

but at least we have a picture of how these different areas together so uh that's it thank you very much for participating in this class uh Andrew will be doing your questions sorry um could you just repeat the last like 30 seconds from andrew will be Chris will be last discussion session on Saturday the 28th of October at the usual time 8 AM Pacific time and the interactive Q&A is open I really appreciate your questions I try to answer them uh and thank you very much awesome can I ask a question or do you want to call it here sure go ahead all right inks wrote Chris spoke last time introducing the mechanisms of scale transition um I'll read your question then I'll fix the cropping on the live stream Chris spoke last time introducing the mechanisms of scale transition of these essentially semantic theories about how one scale embeds in another scale I would love to learn more about this especially in terms of natural language semantics and human cognition semantics versus machine learning wow well that's that's a very good question um I think that's a very relevant question given uh the sorts of machine learning that we see deployed in uh systems like chat GPT and especially as these systems start to associate language with images for example and as such systems are embedded in uh robotic systems that can actually move around in the world and interact with real objects I think one of the one of the key questions in natural language semantics is to what extent does it depend on reference uh to what extent do the meanings we attach to words um or the usages that we uh allow for Words depend on our ability to connect words to to external objects and to what extent uh do those abilities uh rely on connections between words themselves so if you think of of an llm like chat GPT by itself it's just a statistical model of relationships between words as used in some Corpus of text which is large of course but uh we can debate its representativeness and so that's a model of uh semantics that's entirely self-contained it's entirely relations between different pieces of language whereas when we start to connect those words statistics to images or better yet to actual objects that are interacted with uh not just visually but haptically for example by a robot they start to be connected to reference and that transition I think is going to be very very interesting to watch and it's just starting so we all have ringside seats on an incredible experiment in semantics and um so so that would be my response really to that question is is keep your eyes on experiment and see how it goes because it's bound to tell us something about uh how semantics really works and I suspect that there will be enormous opportunity to start relating the results of that experiment to uh what we see in in developmental psychology for example wow awesome all right Tucker asks areas of macroscopic quantum mediated phenomena in biology that get Chris Fields excited um I think the one that excites me the most right now is just this question of how cells

are managing to do the information processing that they're doing and um how much of it they can accomplish or I think the right way to put it is how can they get away with so little classical information and um I don't think we have the cell biology yet to answer that question but I think asking the question may help us design the right kinds of Cell Biology to look for an answer and our cell biology has been organized around the idea that the structures that we see are essentially static that they're classical and that they're constructed in a way that's thermodynamically irreversible but if we think of what we measure as not necessarily reflecting a static structure uh I think we'll start to uh conceptualize it differently and build models of it differently I mean clearly when we look at things like uh protein structure we're looking at snapshots of something that's very Dynamic and when we look in a chemistry textbook and we see you know pictures of this conformation that changes to that conformation we think of those as static stable structures that's kind of a switch uh but it's not actually a switch the way a light switch is it's a much more Dynamic process and the kinds of molecular Dynamic simulations that allow you to study that process are incredibly complicated and and time expensive and and uncertain uh due to resolution problems and and other things but uh I think thinking of these as as kind of fluid reversible processes um is going to change the way we think about them that's a great point and it reminds me of the more physics based way of thinking about the paths that could happen between observations or the statistics based way of samples of a dynamical process and in biology we know that even if we take a snapshot okay this organism is 1 centimeter today it's 2 centimeters tomorrow something changed so we know there was a developmental process then we know there's an ecological process and then there's an evolutionary context Eco evo and then from that more biological understanding of contextual developing organisms all of these different biological phenomena actually help us expand our space of measurement and Analysis for material that on one hand some people may have called inanimate but then they have their own Dynamics different time scales and so bringing some of those like the challenges of measuring and observing intervening in historical biological things gives us a bigger understanding even on measuring and summarizing analyzing ensembles or scales that that don't immediately appear to have that kind of uh but in one sense they clearly are and uh to the extent that we can think of these structures that are much larger than we are individually as living systems cognitive systems Etc I think we'll understand their Dynamics better okay final question upcycle Club wrote How can a Quantum system detect and respond to Quantum context shifts switches which are changes in the measurement basis or the entanglement structure of the system um well uh the

an the answer here is is simple and not very informative it's by making measurements which is all that Quantum systems can actually do and um making measurements that follow probing actions that uh try to push the boundaries of the model of a context uh you know when when having this discussion I I always like to go back to the frame problem in artificial intelligence which is the problem um formulated in the late 60s of predicting what what won't change is the result of an action so it's sort of the inverse of the problem of predicting the side effects and if you can predict all of the side effects You can predict what doesn't change so it's equivalent to the the question of predicting all the side effects and context change is a side effect and it's often in unobserved and maybe unobservable side effect but if you have a model that says I'm working in this context then you can intentionally probe the boundaries of that model and see if you can actually alter the context you're in as a way of discovering what context you're in and know humans do that in Social settings all the um but I I I I think this is something that we need to uh to think about happening at many different scales I suspect that cells are doing this all the time we just don't know how there's so much to say there it just makes me think of the the Paradox or the tension between expecting the expected expecting the unexpected and life needing to expect both of that right so thank you Chris it's been an amazing lecture series many people in the live chat are very appreciative and and um we'll look forward to their submitted questions and their participation in the final discussion coming soon so thank you Chris till next time okay and thank you all bye ciao